

Mama Budget: Research-Informed Simulation

50 rule-based synthetic profiles × 50 purchases = 2,500 simulated labels

These are synthetic votes—not real mums/aunts, not observed research, and not a representative estimate. Use only for dry-running rules and exposing edge cases.

What “research-informed” means here

The sources guide the **dimensions** and guardrails, not the vote percentages. A U.S. study of 2,322 undergraduates linked perceived parental influence over money learning with more responsible spending, but found spending measures did not function uniformly across all sampled U.S. regions. That supports deliberately varied decision lenses; it does not say how a mother would classify any purchase. A Federal Student Aid handbook lists food/housing, required books and program equipment, qualifying fees/insurance, and certain transportation costs in student cost-of-attendance categories, subject to conditions and individual circumstances. These are need-context anchors, not a universal moral taxonomy. A 2015 qualitative Nigerian migrant study in Benin City documented remittance-related family support alongside demands, expectations, and conflict; it motivates a context-sensitive family-transfer label rather than assuming every transfer is automatically NEED or WANT.

Simulation design

Balanced lenses. 50 rule-based synthetic raters are split equally across five hand-designed lenses: essentials-first, budget-guarding, context-first, family-care priority, and wellbeing-balanced. Equal counts are a stress-test design choice, not an estimate of how common these views are.

Three labels. NEED anchors cover clearly stated necessities; WANT anchors cover clear duplication/impulse; DEPENDS captures missing budget or personal context, social/wellbeing choices, upgrades, family transfers, and mixed purchases. Every vote is generated by transparent fixed rules in the workbook.

Mixed baskets. Items 47 and 48 combine necessary and discretionary products. Their synthetic labels are DEPENDS to expose a product problem: split a basket into line items before judging it.

Training boundary. This file can exercise the aggregation code and help identify edge cases. It must not be described as 50 mothers' answers or used as ground truth for a model claiming to represent mothers. For the judging benchmark, collect consented real labels, keep disagreement visible, and evaluate frozen rules on held-out cases.

Simulated output (synthetic only)

Case / purchase	Votes: Need / Want / Depends	Context behind simulated judgment
36 · Chipotle, \$13	10 / 8 / 32	Eating is a need; choosing restaurant food at this price is context-dependent.
23 · Concert ticket, \$85, Ticketmaster	0 / 10 / 40	Social connection competes with cost and budget context.
30 · Nails, \$60, nail salon	0 / 10 / 40	Personal care norms, income, and work/social context vary.
31 · Weekend trip to Chicago, \$180, Greyhound	0 / 10 / 40	Connection and recreation versus budget; no budget balance supplied.
39 · Birthday gift for mum, \$40, Amazon	0 / 10 / 40	Relationship value versus budget is personal; not automatically waste.
40 · Jollof and chicken, \$18, Nigerian restaurant	0 / 10 / 40	Food choice plus belonging/comfort; budget and alternatives are unknown.
42 · Haircut, \$25, barber	0 / 10 / 40	Personal-care frequency and requirements vary.
49 · Money sent home to family, \$50, Remitly	10 / 0 / 40	Remittances may support family basics and carry obligations, but affordability and family circumstances differ.

Sources

- **LeBaron-Black et al. (2023), Journal of Financial Counseling and Planning.** “Financial Socialization Agents and Spending Behavior of Emerging Adults.” N=2,322 undergraduates from three U.S. regions; the study concerns students’ reports of influences/spending, not mothers’ item labels. DOI.
- **U.S. Federal Student Aid Handbook 2025–26.** “Cost of Attendance (Budget).” Lists relevant allowable student budget categories and conditions. Official handbook.
- **Ikuomola (2015), SAGE Open.** “Unintended Consequences of Remittance: Nigerian Migrants and Intra-Household Conflicts.” Exploratory qualitative interview study; it is not representative of all Nigerian families. DOI.